CLIMITIGATE

An AI-Driven IoT-Based System for Real-Time Environmental Monitoring and Climate Risk Mitigation

TEAM BUKTU

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MEET THE TEAM

Christofer D. Banluta

Web Development overall Monitoring Handled the design and development of the web and mobile applications using web technologies. He also oversaw the system integration and ensured smooth data visualization and monitoring.

Jeter Kenzi T. Corpus

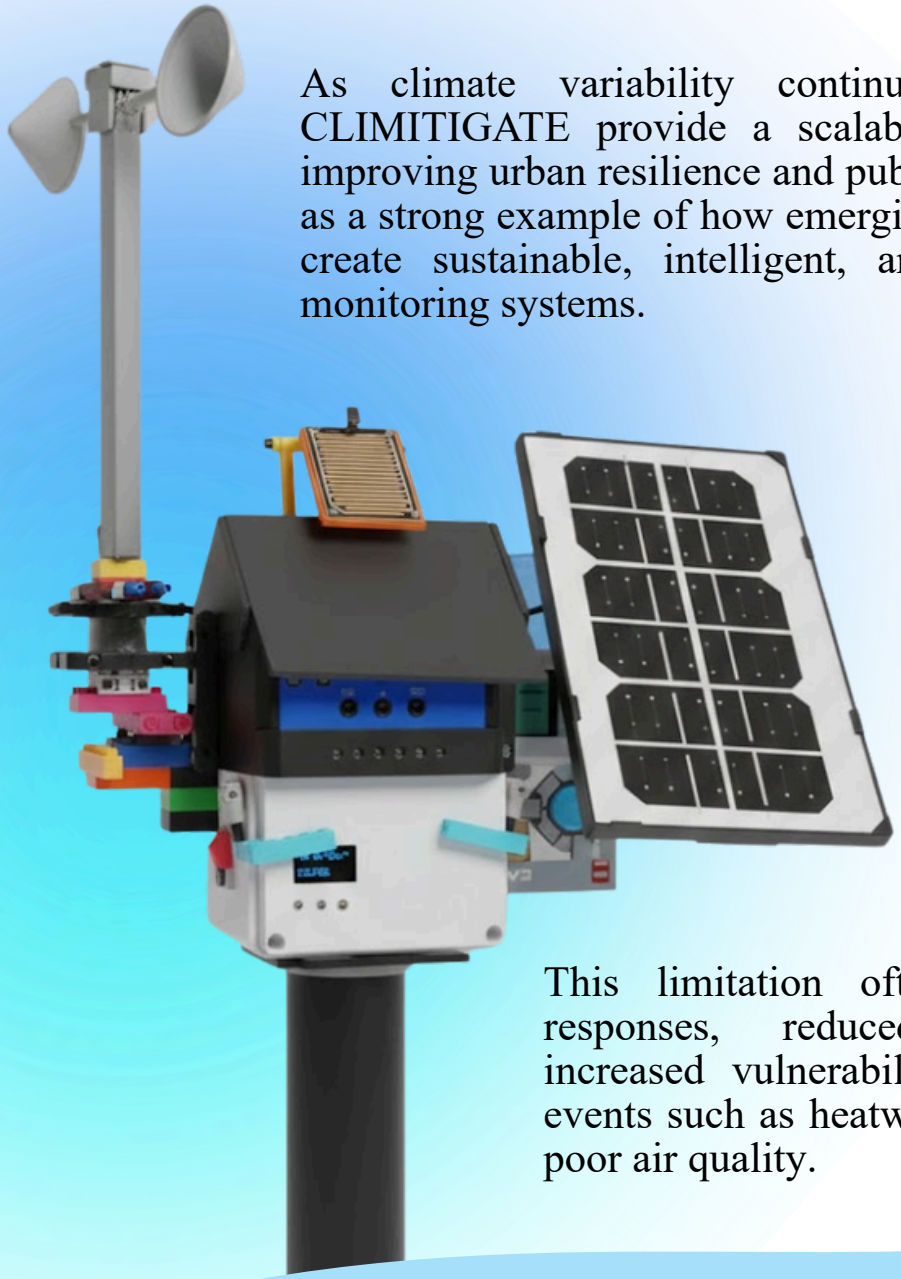
System Programming Responsible for the system logic, microcontroller programming, and data handling. He developed the core functions that allow the ESP32 to collect, process, and transmit environmental data efficiently.

Paulyn Anne C. Timpog

Documentation, Project Design, and Structure Responsible for research documentation, system design planning, and project organization.

PROJECT OVERVIEW

CLIMITIGATE demonstrates the powerful synergy between IoT, artificial intelligence, and cloud computing in addressing real-world environmental challenges. By transforming raw sensor data into meaningful, actionable insights, the system empowers individuals and communities to make informed decisions regarding health, safety, and environmental awareness.



As climate variability continues to intensify, solutions like CLIMITIGATE provide a scalable and cost-effective approach to improving urban resilience and public preparedness. The project stands as a strong example of how emerging technologies can be harnessed to create sustainable, intelligent, and human-centered environmental monitoring systems.

In today's era of rapid climate change, accurate and localized environmental monitoring has become essential for public health and disaster preparedness. However, most traditional weather systems only provide generalized data that do not reflect the actual environmental conditions experienced by specific communities.

This limitation often results in delayed responses, reduced preparedness, and increased vulnerability to extreme weather events such as heatwaves, heavy rainfall, and poor air quality.

PROJECT OVERVIEW

What makes CLIMITIGATE unique is the integration of the Gemini AI model. The system does not only display data but also analyzes patterns and trends using machine learning techniques. It generates visual outputs such as charts and heatmaps, allowing users to easily understand environmental conditions.

Studies have shown that access to localized and real-time weather information significantly improves public safety and reduces health risks associated with climate-related hazards (Smith et al., 2023; World Health Organization, 2024).

OBJECTIVES:

- Provide real-time and localized environmental data
- Improve public awareness and disaster preparedness
- Support health and safety decision-making
- Use artificial intelligence and IoT technology for accurate monitoring
- Transform raw environmental data into meaningful and actionable insights

KEY TECHNOLOGIES USED:

- ESP32 Microcontroller – Data collection and transmission
- Environmental Sensors – Temperature, rainfall, wind, air quality
- Firebase – Cloud data storage and synchronization
- HTML, CSS, JavaScript – Web application
- Gemini AI Model – Data analysis and prediction
- Machine Learning Algorithms – Forecasting and trend analysis

MECHATRONICS SOLUTION

CLIMITIGATE is an AI-powered weather monitoring system driven by an ESP32-S2 microcontroller and an integrated sensor suite, including a BMP180 for barometric pressure, a DHT11 for temperature and humidity, an anemometer for wind speed, and a rain sensor for precipitation intensity.

Programmed in C++ via the Arduino IDE, the system displays local readings on an SSD1306 OLED screen before transferring the data to a Firebase Cloud Database. There, the Google Gemini AI model analyzes historical and real-time data to provide intelligent insights.

These findings are presented through a web application built with HTML, CSS, and JavaScript, which utilizes machine learning to generate visual analytics, such as charts and heatmaps, that help users clearly understand environmental patterns and conditions.

RESULTS

The performance of CLIMITIGATE was evaluated by comparing its outputs with traditional weather monitoring systems following the standards of the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAG-ASA). Unlike conventional systems that provide regional-scale data, CLIMITIGATE collects localized and real-time environmental information, allowing more accurate monitoring of temperature, rainfall, wind speed, and air quality at the community level.

Results showed that CLIMITIGATE's sensor readings were within acceptable ranges based on PAG-ASA reference data, demonstrating that the system delivers reliable results despite using low-cost IoT components. This confirms that CLIMITIGATE effectively complements traditional weather monitoring by providing more precise and location-specific environmental data.

The inclusion of the Gemini AI model allowed the system to analyze trends, predict possible conditions, and provide user-friendly outputs such as visual graphs and health-related suggestions. This innovation made the system more accessible to non-expert users and increased its practical value in real-life situations.

PROTOTYPE TESTING

Before data collection, the CLIMITIGATE system underwent calibration and testing to ensure sensor accuracy. Each sensor was calibrated through a trial-and-error process to minimize fluctuations and stabilize readings. Once properly calibrated, the system's measurements were compared with traditional weather monitoring devices commonly used for environmental data collection, following the standards set by the PAG-ASA particularly for rainfall intensity and weather measurements.

One of the main challenges encountered was calibrating the anemometer, as wind speed had to be accurately computed from voltage readings. Multiple adjustments were required to achieve acceptable accuracy. After thorough calibration, the system produced reliable results that closely matched standard measurements, confirming that CLIMITIGATE is capable of delivering accurate and dependable environmental data despite using low-cost IoT components.

IMPACT

The CLIMITIGATE system transforms environmental data from a passive log into an active decision-support tool. By providing context-aware recommendations such as attire suggestions and health precautions, the system empowers individuals to mitigate the risks of heatwaves and respiratory irritants. On a broader scale, the project offers a scalable solution for urban resilience, demonstrating that the fusion of low-cost hardware and generative AI can democratize access to hyper-local, life-saving information.

FUTURE IMPROVEMENT

To evolve CLIMITIGATE into a more robust and autonomous infrastructure, the following enhancements are proposed:

- Hardware & Precision: Integration of higher-precision industrial-grade sensors to improve the granularity of data collection.
- Actual Anemometer Integration instead of a makeshift for a better performance.
- Geospatial Intelligence: Addition of GPS-based environmental mapping to visualize data trends across specific coordinates.
- Enhanced Connectivity: Development of automated alert systems via SMS to reach users in areas with poor internet connectivity.
- Scalability: Expansion to multi-location monitoring networks to create a mesh of data points across a city.
- Public Governance: Development of a government-accessible dashboard to assist local authorities in disaster response and urban planning.